SOFTWARE REQUIREMENT SPECIFICATION

VEHICLE RENTAL AND BOOKING SYSTEM

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Appendix A: Glossary

1.Introduction

Vehicle Rental and Booking System (VRBS) version 1.0 is a standalone desktop application developed using Java Swing with Oracle 21c database. It enables users to rent vehicles (e-bikes, e-cars, normal bikes, cars) on hourly basis and book cab services with driver allocation and pickup address specification. The system supports secure user authentication using hashed passwords and provides comprehensive CRUD operations for administrators, including supplier and vehicle service management.

1.1 Purpose

This SRS defines the functional, non-functional, interface, and performance requirements for VRBS v1.0. It serves as the primary reference for developers, testers, documentation writers, and project management. The document ensures all stakeholders understand system capabilities, constraints, and user interactions for this Java Swing + Oracle 21c vehicle rental and booking platform.

1.2 Scope

VRBS provides rental services (hourly rates for e-bikes, e-cars, bikes, cars) and cab booking services with driver assignment. Users can view drivers, rental addresses, and booking status. Admins perform full CRUD on bookings/rentals, manage suppliers (add/remove), and handle currently-on-service vehicles. The standalone application uses normal user id/password authentication with password hashing/rehashing for security. Expected development duration is 1 months.

1.3 Definitions, Acronyms, and Abbreviations.

VRBS	Vehicle rental and booking system
CRUD	Create , retrieve , update, delete
JDBC	Java database connectivity

1.4 References

The references for the above software are as follows:-

i. IEEE Std 830-1998: Software Requirements Specifications
ii. Oracle 21c JDBC Documentation
iii. Java Swing Developer Guide
iv. ATM SRS sample
v. www.google.com

1.5 Overview

Section 1.0 discusses the purpose and scope of the software.

Section 2.0 describes the overall functionalities and constraints of

the software and user characteristics.

Section 3.0 details all the requirements needed to design the software.

2.Overall Description

2. 1 Product Perspective

- ✓ VRBS is a comprehensive standalone vehicle management system replacing manual rental processes.
- ✓ It integrates rental (hourly) and booking (cab service) functionalities with real-time availability tracking, driver assignment, and supplier management.
- ✓ Built for local deployment using Java Swing frontend and Oracle 21c backend via JDBC/SQL

2.2 Product Functions

The major functions that **VRBC** performs are described as follows:-

- User functions: Login (userid/password with hashing), view available vehicles, rent vehicles hourly, book cabs with driver/address, view booking/rental history, see assigned drivers.
- Admin Functions: Full CRUD on bookings/rentals, supplier management (add/remove), vehicle service status management, user management, reports generation.
- Core Features: Offline model, secure authentication, hourly rental calculator, driver-vehicle matching, address validation
- Secure User Authentication:- After the application starts, the display provides the user with a login screen containing userid and password fields. Passwords are stored and retrieved using hashing and rehashing for security. After successful authentication, users access their personalized dashboard.
- Vehicle Type Selection:- The user has the freedom to select vehicle category to which rental applies, i.e., E-Bike, E-Car, Normal Bike, or Car
- Rental Duration:- The software allows the user to select rental duration in hours (minimum 1 hour, maximum 24 hours per session) and if there is a delay in returning the rented vehicle , the user have to pay the penalty
- Multi-Vehicle Cart:- Users can add multiple vehicles (maximum 5) to shopping cart before finalizing rental. Real-time total calculation includes base rates, peak charges, and taxes.
- Pickup Address Entry:- User must specify pickup location (address + lat/long coordinates) for rental confirmation.
- Rental History:- Users view complete rental and booking history with filtering by date range, vehicle type, and payment status.
- Reports Dashboard:- Generate daily/monthly revenue reports, vehicle utilization statistics, late return analytics, top performing suppliers/drivers, and cash collection status.

- Data Export:- Export all reports in PDF/Excel format with professional formatting and company branding.

2.3 User Characteristics

There are different kinds of users that will be interacting with the **Vehicle Rental and Booking System (VRBS)**. The intended users of the software are as follows:-

- User A: Novice Customer ,This user has little or no experience with vehicle rental applications and is not a frequent renter. User A will find the product easy to use due to simple user interface and there wont be any complex process on renting and booking vehicles
- User B: Experienced Renter, This user has rented vehicles multiple times before and prefers quick booking processes. User B has saved pickup addresses, favorite vehicle types, and uses the quick rental cart for multi-vehicle bookings. The system provides minimal help sessions.
- Admin User: Rental Company Staff, He/She is familiar with the complete functioning of VRBS and Oracle 21c backend. The admin is in charge of vehicle inventory management, supplier coordination, driver assignments, This user is presented with a management dashboard when logging in with administrator credentials and is provided with advanced CRUD panels.

2.4 Constraints

The major constraints that the **Vehicle Rental and Booking System (VRBS)** project has are as follows:-

- The number of invalid login attempts must not exceed three. After three unsuccessful login attempts, the user account is locked and needs to be unlocked by admin.

- Simultaneous multi-device access to the same user account is not supported (single desktop deployment only).
- The minimum rental duration is 1 hour and the maximum rental duration per session is 24 hours.
- The maximum number of vehicles per rental session is 5 vehicles (multi-vehicle cart limit).
- Before rental confirmation, a check is performed by the system to ensure vehicle availability and driver availability (for cab bookings) failing which the booking is denied.
- Peak hour pricing (7-10 AM, 5-9 PM) applies 1.5x multiplier to base rates automatically.
- Late return penalty of ₹50 per hour applies after 1-hour grace period.
- Cancellation policy: Full refund within 15 minutes, 50% fee after 15 minutes.
- Users can only rent/book vehicles marked as "Available" in the system inventory.

2.5 Assumptions and Dependencies

The requirements stated in the **Vehicle Rental and Booking System (VRBS)** SRS could be affected by the following factors:

- One major dependency that the project might face is the changes in rental pricing policies, peak hour definitions, and penalty structures. As the rental company updates its pricing strategy or government regulations change (electric vehicle subsidies, emission norms), the system needs to be updated immediately through admin rate configuration panels. A delay in updating VEHICLE_RATES and PEAK_HOURS tables will result in

revenue loss or customer disputes. So these changes should be configurable by admin as and when required by the developer using Eclipse IDE.

- Another constraint relating to the operating environment is that we are specific to Oracle 21c Express Edition with JDBC Thin Driver (ojdbc11.jar). The system will not function with MySQL, PostgreSQL, or other databases without complete DAO layer rewrite.

3.External Requirements

3.1.1 User Interface Requirements

The interface provided to the user should be a very user-friendly one and it should provide an optional interactive help for each of the service listed. The interface provided is a menu driven one and the following screens will be provided:-

1. A login screen is provided in the beginning for entering the required username/pin no. and account number.
2. An unsuccessful login leads to a reattempt (maximum three) screen for again entering the same information. The successful login leads to a screen displaying a list of supported languages from which a user can select any one.
3. After the login, a screen with a number of options is then shown to the user. It contains all the options along with their brief description to enable the user to understand their functioning and select the proper option.
4. A screen will be provided for user to check his account balance.
5. A Rental History screen provided for user to check past rentals and bookings with total spent, vehicle types, and payment status.

The following reports will be generated after each session dealt with in the system:-

1. The login time and logout time along with user's ID and role is registered in AUDIT_LOG table.
2. The application instance ID (localhost deployment) through which session established is noted in database.
3. Rental/booking transactions, payment details, late penalties are recorded in HISTORY tables.
4. A digital receipt (PDF) is generated for user displaying rental/booking details, driver info, total amount..

Other various user interface requirements that need to be fulfilled are as follows:-

- Java Swing screens minimum 1024x768 resolution
- Tabbed interface with JTabbedPane for Rental/Booking/Admin
- Interactive tooltips on all buttons and fields
- Progress bars for booking confirmation and report generation
- Confirmation dialogs for destructive actions (cancel, delete)
- JTable for vehicle lists, history, admin CRUD operations
- Search/filter functionality on all list screens
- Responsive design adapts to window resize
- PDF export buttons on all report screen

3.1.2 Hardware Interface Requirements:

There are various hardware components with which the **Vehicle Rental and Booking System (VRBS)** is required to interact. Various hardware interface

requirements that need to be fulfilled for successful functioning of the software are as follows :

- Power supply: Standard 220V AC with UPS backup (minimum 30 minutes)
- RAM: Minimum 2GB (Oracle 21c + Java Swing)
- Processor: Minimum Intel i3 or equivalent
- Storage: 50GB SSD (Oracle database + application files)

3.1.3 Software Interface Requirements

In order to perform various different functions, this Vehicle Rental and Booking System (VRBS) software needs to interact with various other softwares. So there are certain software interface requirements that need to be fulfilled which are listed as follows:-

- The transaction management software used to manage rental/booking transactions and keep track of vehicle availability shall be Oracle 21c PL/SQL procedures.
- The user authentication software used to verify userid/password (BCrypt hashed) and login shall be Java Swing + JDBC authentication module.
- The pricing calculation engine for peak/off-peak rates and late penalties shall be VEHICLE_RATES table + PL/SQL functions.
- The database used to keep record of users, vehicles, drivers, bookings, rentals, suppliers shall be Oracle 21c Express Edition.
- The GUI framework for all desktop screens shall be Java Swing 17 with JDBC Thin Driver (ojdbc11.jar).
- The IDE used for development and deployment shall be Eclipse 2023-12+

3.1.4 Communication Interface Requirements

The VRBS desktop application needs to communicate with localhost Oracle 21c database for each session for various functions such as login verification, vehicle availability check, booking confirmation, payment processing etc. so the

following are the various communication interface requirements that are needed to be fulfilled in order to run the software successfully:-

- The system will employ JDBC Thin Driver with localhost connection (127.0.0.1:1521) for zero-cost communication.
- The communication protocol used shall be Oracle JDBC Protocol over TCP/IP localhost.

4. System Features

4.1 Vehicle Rental and Booking Management

Description

The system is designed to provide the user with the facility of vehicle rental and cab booking services through a Java Swing desktop interface without the aid of rental company staff. The functioning of the system shall be as follows:-

At the start, the user is provided with a login screen and is required to enter userid and password (BCrypt hashed) which are then verified against Oracle 21c USERS table.

In case of an unsuccessful attempt, user is asked again for credentials but the maximum number of attempts given is limited to 3 only, failing which account is locked and needs to be unlocked by admin for future use.

After a successful login, user is directed to role-based dashboard:

- Normal User: Rental tab, Booking tab, History tab
- Admin User: Management dashboard with CRUD operations

After login, user is presented with main dashboard that displays Rental Services, Cab Booking, My History with large icon buttons and brief descriptions enabling user to understand each service functioning.

The system provides user with vehicle rental services such as:

- Hourly rentals for E-Bike, E-Car, Normal Bike, Car
- Peak/off-peak pricing automatically applied (7-10AM, 5-9PM × 1.5)
- Multi-vehicle cart (maximum 5 vehicles per session)

- Pickup address entry with lat/long coordinates

The system also provides cab booking services where:

- User enters pickup location and destination
- System auto-assigns nearest driver using Euclidean distance algorithm
- User views driver details (name, photo, rating, contact, vehicle)
- Real-time booking status: Confirmed → Assigned → Arrived → Completed

The rental process works as follows:

1. User selects vehicle type from availability grid
2. Enters rental hours (1-24 hours)
3. System shows real-time pricing (base + peak multiplier)
4. User adds to cart (multiple vehicles supported)
5. Confirms pickup address and completes payment
6. Digital receipt generated with QR code and expected return time

At any moment if user wants to abort transaction, Cancel button cancels all cart items and returns to dashboard. User can start fresh rental/booking.

After user finishes rental/booking, for security purpose, session automatically times out after 15 minutes inactivity or user clicks Logout button.

Validity Checks

In order to gain access to the system, user must enter correct userid/password failing which account locks after 3 attempts.

User can access only one account at a time with their own credentials.

Only available vehicles (status = 'AVAILABLE') can be rented/booked.

Peak pricing automatically enforced during 7-10AM, 5-9PM.

Minimum 1 hour rental, maximum 24 hours per session.

Maximum 5 vehicles per cart.

Admin requires separate admin credentials with elevated privileges.

Sequencing Information

The information about the users and their account should be entered into the

database prior to any of the transactions and the backup be maintained for all account information

Error Handling/Response to Abnormal Situations

If any of the above validation/sequencing flow does not hold true, appropriate error messages will be prompted to the user through Java Swing JOptionPane dialogs for doing the needful

5. Other Non-functional Requirements

5.1 Performance Requirements

The following list provides a brief summary of the performance requirements for the Vehicle Rental and Booking System (VRBS) software:

5.1.1 Capacity

The VRBS shall provide 24 hour service for rental and booking operations on localhost Oracle 21c deployment supporting maximum 50 concurrent users.

5.1.2 Dynamic Requirements

- Login verification time must not exceed 1 second under normal workload and 1.5 seconds under peak workload (50 users)
- Vehicle availability check must not exceed 0.5 seconds under normal workload and 1 second under peak workload
- Rental/booking confirmation time must not exceed 2 seconds under normal workload and 3 seconds under peak workload

5.1.3 Quality

The primary objective is to produce quality software. As the quality of a piece of software is difficult to measure quantitatively, the following guidelines will be used when judging the quality of the software:

1. Consistency – All Java Swing screens follow uniform color scheme (blue/white theme), font sizes (Arial 12pt), and button layouts.
2. Test Cases – All functionalities are thoroughly tested

5.2 Software System Attributes

5.2.1 Reliability

The JDBC communication protocol shall ensure reliability and quality of data transmission between Java Swing frontend and Oracle 21c localhost database.
The Oracle 21c database shall use non-volatile SSD storage with automatic transaction logging.
All rental/booking transactions use ACID properties with rollback on failure.
Automatic connection retry (3 attempts) for temporary JDBC disconnects.

5.2.2 Availability

The product desktop will have UPS backup power supply (minimum 30 minutes) in case of power failures.
Any abnormal operations (JDBC deadlock, out-of-memory) shall log error and gracefully restart application.
After abnormal shutdown of VRBS, the system shall auto-recover on next login using last consistent database state.
No data inconsistency introduced during transaction failure due to Oracle transaction rollback.

5.2.3 Security

The system shall implement BCrypt password hashing with automatic rehashing for all user credentials.
The system shall have two levels of security:

1. User authentication (userid + hashed password)
2. Role-based access control (User vs Admin privileges)

The password shall be hashed while storing and the password shall be 8-20 characters long with one capital character , combo of small characters and numbers.

5.2.4 Maintainability

The system components i.e. Java Swing frontend, Oracle 21c database, JDBC drivers, configuration files shall be easily serviceable without requiring application restart.

The system should have the mechanism of self-monitoring through:

- Daily database health checks (tablespace usage, connection pool)
- JDBC connection monitoring (auto-reconnect on failure)

- Log file analysis (error patterns detection)
- Performance metrics (response time alerts)

6. Other Requirements

None

Appendix A: Glossary

CRUD	Create, Read, Update, Delete operations
DAO	Data Access Object - Java classes for database operations
JDBC	Java Database Connectivity for Oracle 21c connection
JDBC Thin	Lightweight Oracle driver (ojdbc11.jar - 5MB)

JTable	Java Swing component for displaying tabular data
JTabbedPane	Java Swing component for tabbed interface
MVC	Model-View-Controller architecture pattern
Oracle 21c	Database system (Express Edition - free for localhost)
PL/SQL	Oracle procedural language for stored procedures
PreparedStatement	JDBC class preventing SQL injection attacks
SQL*Plus	Oracle command-line interface for backups
Swing	Java GUI framework for desktop applications
ACID	Atomicity, Consistency, Isolation, Durability (transaction properties)